Artificial Intelligence (AI) is the simulation of human intelligence processes by computer systems. These processes include learning, reasoning, problem-solving, perception, and language understanding. John McCarthy coined the term Artificial Intelligence in 1956 during the Dartmouth Conference.

Search algorithms form the cornerstone of classical AI problem solving. An uninformed search algorithm, also known as blind search, explores the search space without domain-specific knowledge beyond the problem definition. Breadth-First Search (BFS) is an uninformed search algorithm that expands the shallowest nodes first using a FIFO queue. BFS is complete and optimal when step costs are identical. In contrast, Depth-First Search (DFS) explores the deepest unexpanded node using a LIFO stack. DFS has lower memory requirements than BFS, but it is neither complete in infinite state spaces nor optimal.

Informed search algorithms utilize heuristic evaluation functions to estimate the cost of reaching the goal. The A* search algorithm is a widely used heuristic search algorithm that computes f(n) = g(n) + h(n), where g(n) is the exact cost from the start node to node n, and h(n) is the estimated cost from node n to the goal. When the heuristic function is admissible, meaning it never overestimates the true cost, A* search is guaranteed to be optimal. Consistent heuristics ensure that the estimated cost between adjacent nodes satisfies the triangle inequality.

Adversarial search applies to competitive multi-agent environments such as chess and checkers. The Minimax algorithm evaluates game trees by assuming optimal play from both the maximizing player and the minimizing opponent. Alpha-Beta Pruning is an optimization technique for Minimax that eliminates branches that cannot influence the final decision. Alpha-Beta pruning significantly reduces the branching factor, allowing search algorithms to evaluate deeper levels within the same time limit.